



My Exploratory Data Analysis by Awotidebe Oluwaseyifunmi (co-written with Samrawit Mewa)

1. Project Overview

- **Project Title:** Data Analysis For Predicting dengue fever outbreaks in Sri Lanka Using Python
- **Goal of the project:** To uncover useful insights through analysis using python
- **Dataset(s) used:** Weather dataset merged with Number of weekly reported cases dataset and Population dataset

2. Data Overview

- **Brief Description of the Data:** The analysis focuses on two datasets covering the period from 2007 to 2023 for 25 districts. The first dataset contains weekly reported dengue cases, which is merged with a dataset that includes various climatic variables such as temperature, rainfall, and humidity. Using this dataset, the goal is to uncover potential correlations between climatic factors and the spread of dengue across different regions.

Below is a snapshot of the first five records for the weekly reported dengue cases merged with the climatic variables:

	District	Number_of_Cases	Week_Start_Date	Month	Year	Week	Week_End_Date	Avg Max Temp (°C)	Avg Min Temp (°C)	Avg Apparent Max Temp (°C)	Avg Apparent Min Temp (°C)	Precipitation (mm)	Total Rain (mm)	Avg Wind Speed (km/h)	Max Wind Gusts (km/h)	Weather Code	Avg Daylight Duration (hours)	Avg Sunrise Time	Avg Sunset Time
0	Ampara	0	2007-01-01	1	2007	1	2007-01-08	26.9375	21.8625	28.8500	24.1000	8.0	8.0	18.2000	44.3	3	11.714583	376	1079
1	Ampara	0	2007-01-08	1	2007	2	2007-01-15	26.3500	23.3250	30.2625	26.7000	86.1	86.1	17.0250	43.6	53	11.733333	378	1082
2	Ampara	0	2007-01-15	1	2007	3	2007-01-22	27.2125	23.1625	29.5750	25.9625	3.1	3.1	20.1875	48.2	51	11.754167	380	1085
3	Ampara	0	2007-01-22	1	2007	4	2007-01-29	26.7250	23.0375	29.9500	26.1125	53.6	53.6	17.2000	42.5	61	11.783333	381	1088
4	Ampara	0	2007-01-29	1	2007	5	2007-02-05	27.3625	22.9625	30.7500	26.1250	8.2	8.2	17.2625	40.0	51	11.812500	382	1090

✓ Weather_report/Reported Cases Column names description

1. `District` - The Administrative districts in Sri Lanka
2. `Number_of_Cases` - Dengue cases reported weekly
3. `Week_Start_Date` - The Starting date for the reported cases
4. `Month` - The Month the cases were reported
5. `Year` - The Year the cases were reported
6. `Week` - The Week the cases were reported
7. `Week_End_Date` - The Ending date for the reported cases
8. `Avg Max Temp (°C)` - The Average maximum temperature measured in degree Celsius
9. `Avg Min Temp (°C)` - The Average minimum temperature measured in degree Celsius
10. `Avg Apparent Max Temp (°C)` - The Average Apparent maximum temperature measured in degree Celsius
11. `Avg Apparent Min Temp (°C)` - The Average Apparent minimum temperature measured in degree Celsius
12. `Total Precipitation (mm)` - The Total Precipitation measured in millimetres
13. `Total Rain (mm)` - The Total Precipitation measured in millimetres
14. `Avg Wind Speed (km/h)` - The Average wind speed measured in kilometre per hour
15. `Max Wind Gusts (km/h)` - The Maximum wind gust speed measured in kilometre per hour
16. `Weather Code` - Code representing weather conditions
17. `Avg Daylight Duration (hours)` - The Average daylight duration measured in hours
18. `Avg Sunrise Time` - The Average Sunrise Time
19. `Avg Sunset Time` - The Average Sunset Time

The second dataset provides population data for the same 25 districts, covering the years 2010 to 2023. To extend the dataset back to 2007, linear regression was applied to estimate the population for the years 2007 to 2009. By merging this population data with the weekly dengue cases, the analysis can explore dengue incidence in relation to population changes, providing a more accurate understanding of disease spread and impact across different districts.

This population data, when integrated with the dengue case and climatic variables, will allow for population-adjusted analyses, offering insights into per capita dengue incidence and potential demographic influences on outbreak patterns.



	year	total_population	district
0	2007	2270352	Colombo
1	2008	2284182	Colombo
2	2009	2298012	Colombo
3	2010	2311842	Colombo
4	2011	2325673	Colombo

✓ Population Column names description

1. `Year` - The Year population was counted
2. `total_population` - The population Count
3. `district` - The Administrative districts in Sri Lanka

3. Data Cleaning and Preparation

- **Missing Value Treatment:**

While cross-checking the weekly dengue cases and population datasets, it was found that the district names were not consistently labeled across the two sources. A thorough check was conducted to ensure uniformity in district names, as discrepancies could lead to inaccuracies in the analysis. The necessary adjustments were made to standardize district names, allowing for reliable insights across both datasets.

This step ensures that the dataset accurately reflects the same districts, providing a robust foundation for further analysis.

```
# Change 'Moneragala' to 'Monaragala' in population dataframe

lanka_p['district'] = lanka_p['district'].str.replace('Moneragala', 'Monaragala')
lanka_p['district'].sort_values().unique()

array(['Ampara', 'Anuradhapura', 'Badulla', 'Batticaloa', 'Colombo',
       'Galle', 'Gampaha', 'Hambantota', 'Jaffna', 'Kalutara', 'Kandy',
       'Kegalle', 'Kilinochchi', 'Kurunegala', 'Mannar', 'Matale',
       'Matara', 'Monaragala', 'Mullaitivu', 'NuwaraEliya', 'Polonnaruwa',
       'Puttalam', 'Ratnapura', 'Trincomalee', 'Vavuniya'], dtype=object)
```

3.1 High-Level Analysis of Dengue Cases

Conducting an aggregate analysis of weekly reported dengue cases across all districts provides a broad overview of outbreak trends. This high-level approach summarizes overall patterns and trends without delving into district-specific details.

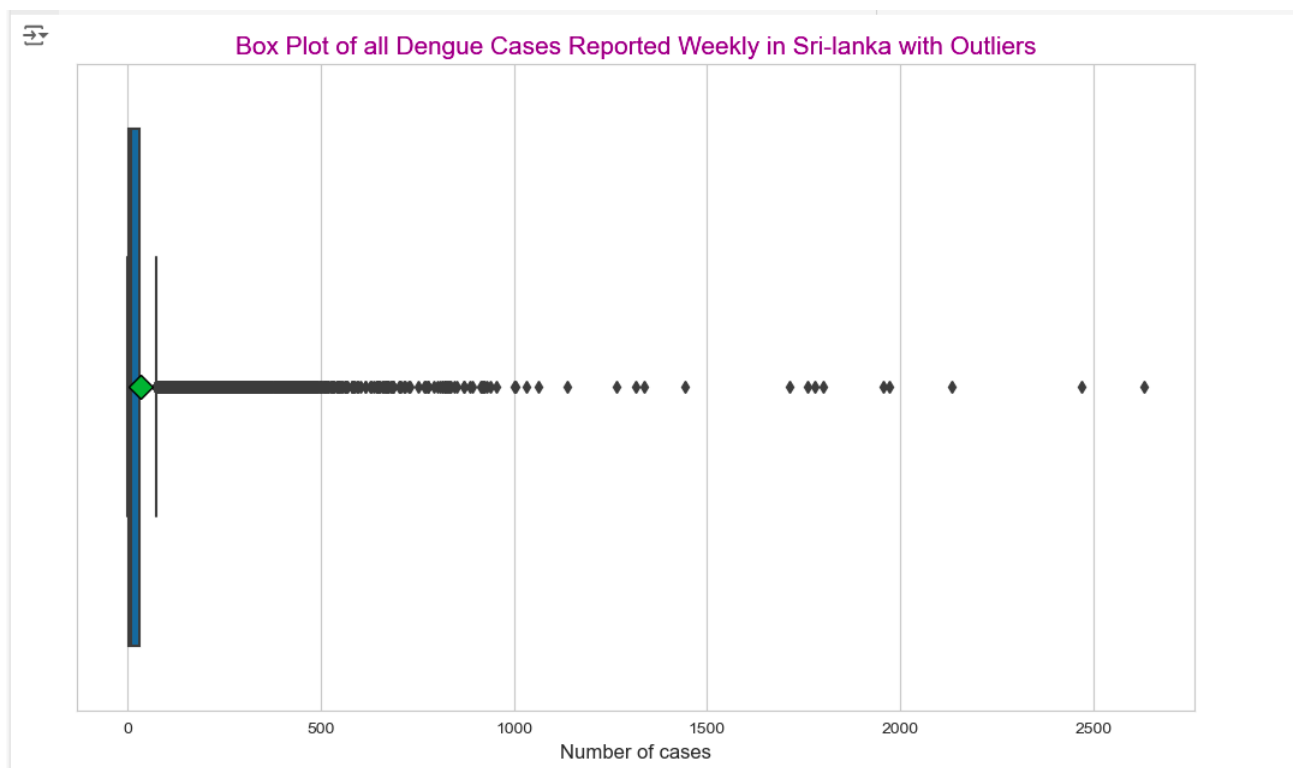
Summary Statistics of Weekly Dengue Cases (High-level analysis)

The overall mean of all weekly reported dengue cases, across all districts, is 35, while the median is 9. These summary statistics provide a quick insight into the distribution of dengue cases, showing a right-skewed distribution where most weeks had relatively fewer cases, but there were periods with significantly higher case counts.

The data is non-negative and not normally distributed, calculating a lower bound for outliers isn't practical. Therefore, only the upper bound is used for outlier detection.

- **Outlier/anomalies Detection & Treatment:**

A box plot of the weekly reported dengue cases, irrespective of district, provides a general view of the distribution and variability over the entire dataset. This visual summary highlights the central tendency, spread, and potential outliers, giving an overall sense of the trends in dengue cases without focusing on specific districts.



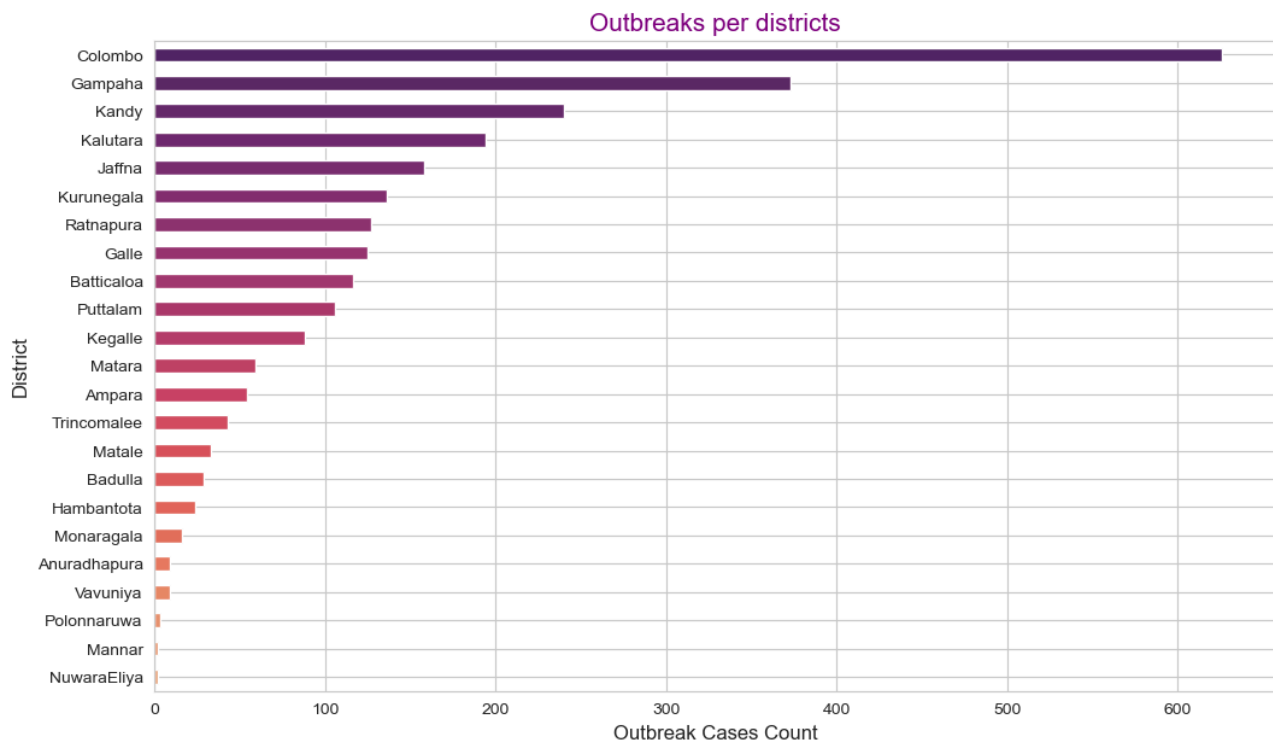
The box plot highlights the distribution of weekly dengue cases across Sri Lanka, showing that most cases fall within the 0 to 74 range. However, there are extreme outliers, with the maximum weekly case count reaching 2631, indicating occasional large outbreaks.

Outbreak Threshold for Weekly Dengue Cases in Sri Lanka

Based on the distribution of weekly reported dengue cases, the typical range is between 0 and 74 cases. Given this, it is reasonable to define an outbreak as any week where the reported cases exceed 74. This threshold helps identify periods of unusually high case numbers, signaling potential outbreaks that require further investigation.

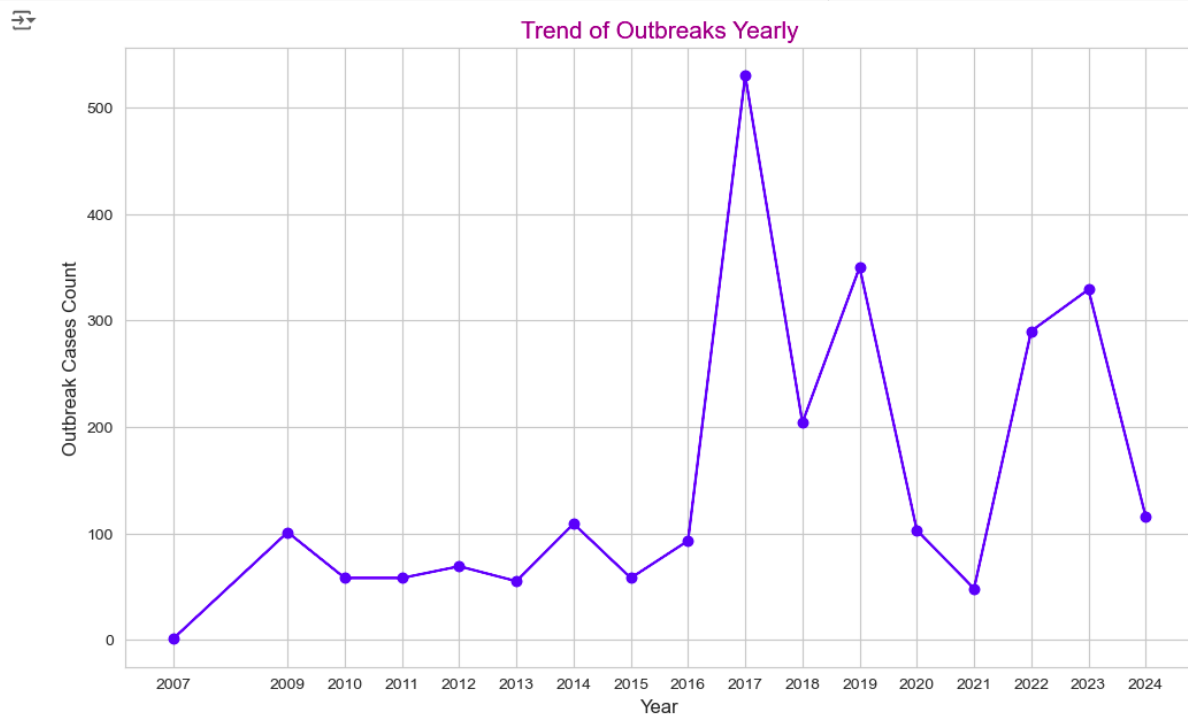
Out of a total of 22,950 weekly dengue cases reported over time, 2,572 cases (11.2%) exceed the outbreak threshold of 74 cases per week. These abnormal cases represent potential outbreak periods, indicating significant surges in dengue incidence that require further investigation.

Outbreaks per districts



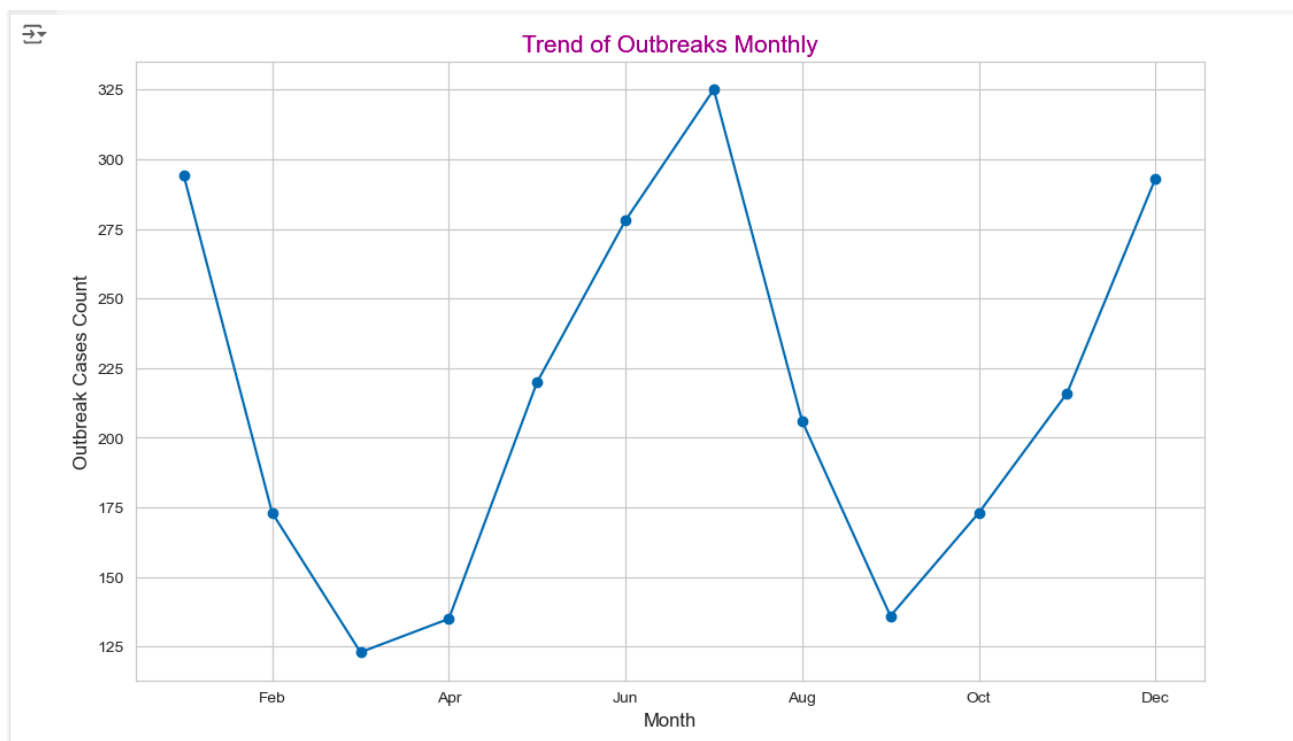
Outbreaks have been observed in 23 out of the 25 districts in Sri Lanka. The districts with the highest number of outbreaks are **Colombo**, **Gampaha**, and **Kandy**, making them the top 3 affected regions. These districts may require more focused public health interventions due to their higher incidence of dengue outbreaks.

Year where these outbreaks occur for Weekly Dengue Cases in Sri Lanka



There was no outbreak of dengue fever in 2008 (which means no weekly cases greater than 74 was reported in 2008), the maximum and minimum number of outbreaks occurred in the year 2017 and 2007 respectively according to the plot above (Trend of Outbreaks Yearly).

Outbreak monthly trend for Weekly Dengue Cases in Sri Lanka



Seasonal Distribution of Dengue Outbreaks in Sri Lanka

Dengue outbreaks in Sri Lanka predominantly occur in July, indicating a peak period for the disease. Conversely, March experiences the fewest outbreaks, suggesting a lower incidence during this month. This seasonal pattern can help in targeting preventive measures and resources more effectively.

The distribution of dengue outbreaks across the year is as follows:

- **January - March:** 590 outbreaks
- **April - June:** 633 outbreaks
- **July - September:** 682 outbreaks
- **October - December:** 667 outbreaks

The highest number of outbreaks occurs from July to September, indicating that dengue activity peaks during the third quarter of the year. In contrast, the first quarter experiences fewer outbreaks relative to the other quarter of the year. This seasonal trend highlights the critical period for increased surveillance and preventive measures in the latter half of the year.

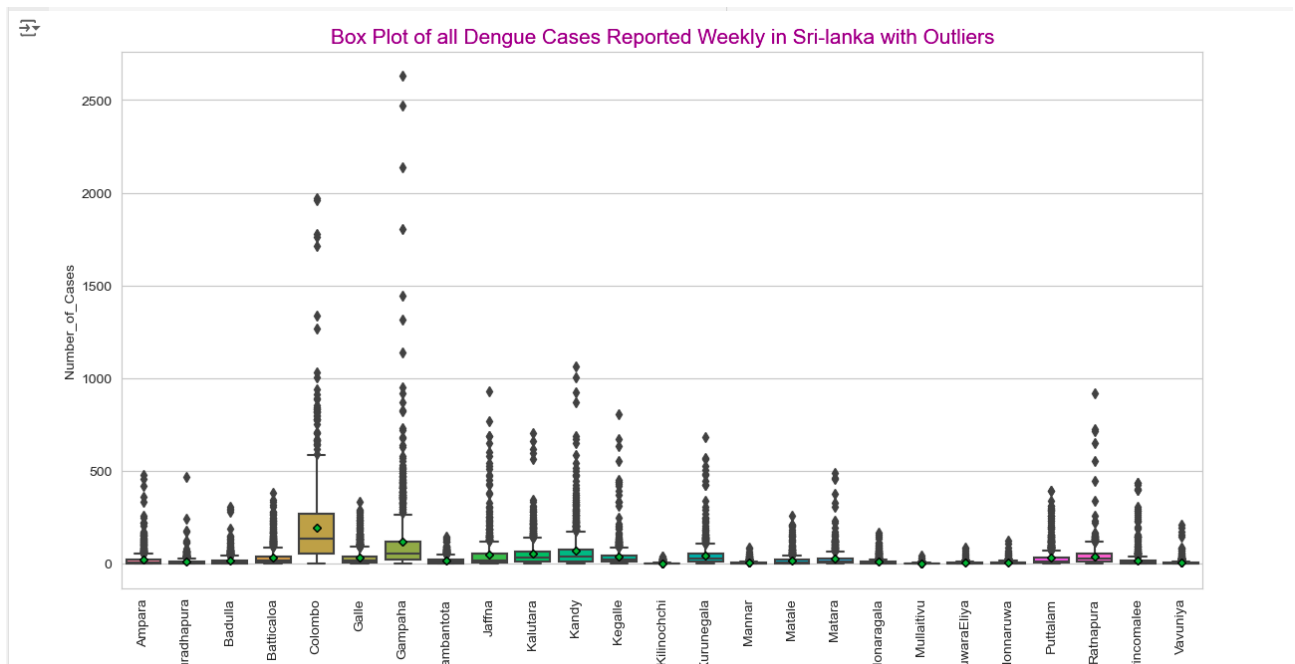
3.2 District-Specific Analysis of Weekly Dengue Cases

Performing a district-specific analysis of weekly dengue cases allows for a detailed examination of outbreaks within each individual district. This approach uncovers unique patterns, trends, and outliers at the district level, providing insights into regional variations and specific local dynamics that might be masked in a broader aggregate analysis.

Weekly Analysis of Dengue Cases by District (Granular analysis)

year	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
district																		
Ampara	8	47	425	484	199	184	401	529	407	478	5689	2048	1693	1275	409	2687	2025	657
Anuradhapura	237	87	560	829	229	307	376	417	252	373	2647	891	994	828	239	499	764	310
Badulla	80	91	297	912	529	240	367	922	361	860	3337	630	1880	488	803	1371	1869	475
Batticaloa	66	73	583	930	1396	569	402	873	1009	414	5057	4895	2699	3990	2951	1263	2977	1147
Colombo	1740	1420	4199	4971	7557	5756	7393	11831	8312	11228	32882	10532	20305	4211	3860	17676	18832	7405
Galle	95	94	624	955	651	730	748	785	669	1846	5417	747	6347	1175	588	3854	3920	1439
Gampaha	779	730	3216	2541	2936	3557	2705	4598	2077	2064	28694	6011	13655	2213	2140	13737	16298	3315
Hambantota	93	131	892	678	333	396	280	517	311	698	3311	1005	1964	363	443	1456	1328	398
Jaffna	117	20	189	1830	327	791	716	1844	1800	2172	5743	4302	8013	2138	414	3579	4351	4034
Kalutara	412	430	864	1194	1048	1259	1454	2201	1230	1879	10126	3267	7000	1653	1581	6346	5173	1753
Kandy	364	320	3080	1232	1513	1329	1291	2055	1075	3100	14090	3885	8855	3385	1083	6588	9326	2615
Kegalle	331	452	2287	694	887	1493	987	1305	593	1019	9999	1576	2621	847	649	3165	3603	1338
Kilinochchi	0	0	0	46	31	24	33	43	40	51	406	349	384	135	56	152	147	280
Kurunegala	680	289	2395	1208	867	2124	1718	2112	933	1840	10899	2510	2995	905	1465	2581	3568	1383
Mannar	5	29	15	541	94	136	53	262	70	158	526	232	255	128	182	335	251	241
Matale	92	185	1866	489	337	360	314	508	280	678	3016	925	2290	556	262	1340	1843	367
Matara	230	354	1040	556	692	1300	450	755	453	1319	6314	1207	3983	444	644	1650	1624	504
Monaragala	52	47	172	784	253	157	169	265	199	357	3196	873	311	0	192	528	694	473
Mullaitivu	0	0	0	11	14	22	85	92	92	150	362	112	162	52	7	68	193	162
NuwaraEliya	36	30	278	201	215	205	214	278	153	388	866	216	426	176	78	206	285	125
Polonnaruwa	61	56	166	340	193	151	359	223	164	304	1421	331	518	246	112	355	630	210
Puttalam	355	241	679	673	377	1055	650	686	406	588	7626	2030	2182	485	670	4256	4061	820
Ratnapura	307	223	1427	1772	757	1712	1110	1924	631	1774	7514	2286	3715	1856	702	3572	3368	1995
Trincomalee	41	157	200	597	142	121	144	520	491	427	4578	1210	2594	2163	271	1147	2146	517
Vavuniya	38	9	1098	538	70	98	53	105	172	190	971	610	749	215	60	129	202	95

Examining weekly reported dengue cases for each district individually may reveal distinct patterns and outliers that are not apparent in aggregate data. This granular analysis can provide more nuanced insights into regional variations, outbreak trends, and specific district-level anomalies, enhancing the overall understanding of dengue dynamics across Sri Lanka.



District-Specific Outbreak Thresholds

Box plot range of reported weekly cases in Ampara is 0 - 52

The Outlier count for Ampara is 85

Box plot range of reported weekly cases in Anuradhapura is 0 - 28

The Outlier count for Anuradhapura is 82

Box plot range of reported weekly cases in Badulla is 0 - 43

The Outlier count for Badulla is 82

Box plot range of reported weekly cases in Batticaloa is 0 - 89

The Outlier count for Batticaloa is 98

Box plot range of reported weekly cases in Colombo is 0 - 588

The Outlier count for Colombo is 42

Box plot range of reported weekly cases in Galle is 0 - 90

The Outlier count for Galle is 102

Box plot range of reported weekly cases in Gampaha is 0 - 266

The Outlier count for Gampaha is 109

Box plot range of reported weekly cases in Hambantota is 0 - 48

The Outlier count for Hambantota is 61

Box plot range of reported weekly cases in Jaffna is 0 - 122

The Outlier count for Jaffna is 78

Box plot range of reported weekly cases in Kalutara is 0 - 140

The Outlier count for Kalutara is 82

Box plot range of reported weekly cases in Kandy is 0 - 176

The Outlier count for Kandy is 92

Box plot range of reported weekly cases in Kegalle is 0 - 89

The Outlier count for Kegalle is 70

Box plot range of reported weekly cases in Kilinochchi is 0 - 8

The Outlier count for Kilinochchi is 76

Box plot range of reported weekly cases in Kurunegala is 0 - 110

The Outlier count for Kurunegala is 61

Box plot range of reported weekly cases in Mannar is 0 - 10

The Outlier count for Mannar is 96

Box plot range of reported weekly cases in Matale is 0 - 46

The Outlier count for Matale is 71

Box plot range of reported weekly cases in Matara is 0 - 66

The Outlier count for Matara is 68

Box plot range of reported weekly cases in Monaragala is 0 - 24

The Outlier count for Monaragala is 75

Box plot range of reported weekly cases in Mullaitivu is 0 - 5

The Outlier count for Mullaitivu is 75

Box plot range of reported weekly cases in NuwaraEliya is 0 - 14

The Outlier count for NuwaraEliya is 56

Box plot range of reported weekly cases in Polonnaruwa is 0 - 18

The Outlier count for Polonnaruwa is 59

Box plot range of reported weekly cases in Puttalam is 0 - 72

The Outlier count for Puttalam is 108

Box plot range of reported weekly cases in Ratnapura is 0 - 119

The Outlier count for Ratnapura is 37

Box plot range of reported weekly cases in Trincomalee is 0 - 38

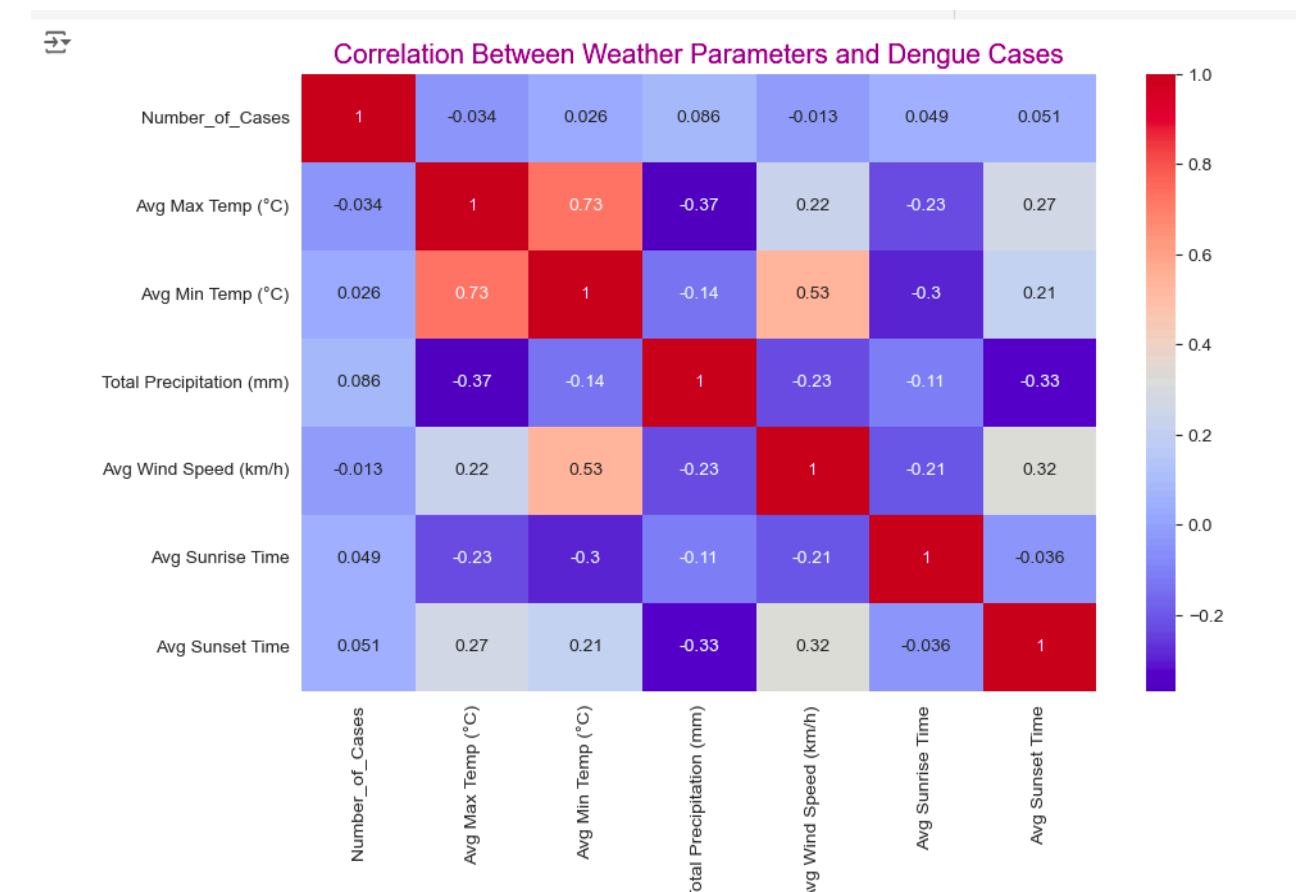
The Outlier count for Trincomalee is 86

Box plot range of reported weekly cases in Vavuniya is 0 - 12

The Outlier count for Vavuniya is 111

The threshold for identifying an outbreak varies by district. For instance, in Vavuniya, an outbreak is considered when the weekly reported cases exceed 12 and the number of outbreaks that has occurred is 111 (Outlier Count). This district-specific range reflects the varying baseline levels of dengue activity across different regions.

4. Exploratory Data Analysis



Correlation Analysis of Dengue Cases with Climatic Variables (High-level)

The relationship between reported dengue cases and various climatic factors was assessed. The following correlations were observed:

- **Average Maximum Temperature (°C):** -0.034041
 - **Correlation:** Very weak negative
 - **Implication:** As the average maximum temperature increases, the number of reported cases shows a slight decrease.
- **Average Minimum Temperature (°C):** 0.026080
 - **Correlation:** Very weak positive
 - **Implication:** As the average minimum temperature increases, the number of reported cases shows a slight increase.
- **Total Precipitation (mm):** 0.085528

- **Correlation:** Weak positive
- **Implication:** Higher total precipitation is weakly associated with an increase in the number of reported cases.
- **Average Wind Speed (km/h):** -0.013081
 - **Correlation:** Very weak negative
 - **Implication:** As the average wind speed increases, the number of reported cases shows a slight decrease.
- **Average Sunrise Time:** 0.049056
 - **Correlation:** Very weak positive
 - **Implication:** An increase in average sunrise time is weakly associated with an increase in the number of reported cases.
- **Average Sunset Time:** 0.050656
 - **Correlation:** Very weak positive
 - **Implication:** An increase in average sunset time is weakly associated with an increase in the number of reported cases.

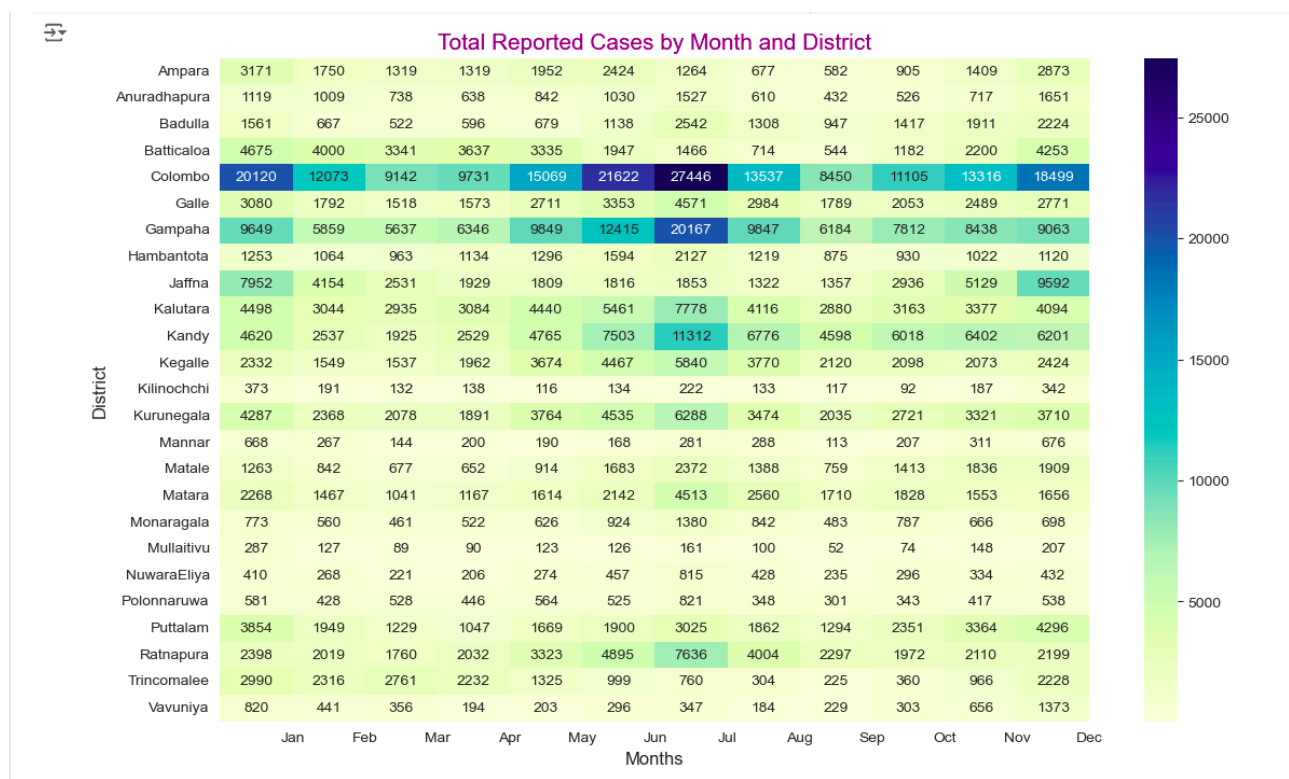
Key Insights

- **Temperature and Wind Speed:** A very weak negative correlation suggests that as maximum temperature and wind speed increase, fewer cases are reported.
- **Precipitation and Temperature:** A weak positive correlation indicates that higher minimum temperatures and total precipitation are associated with more reported cases.
- **Sunrise and Sunset Times:** Very weak positive correlations with the number of reported cases suggest slight increases with longer daylight hours.

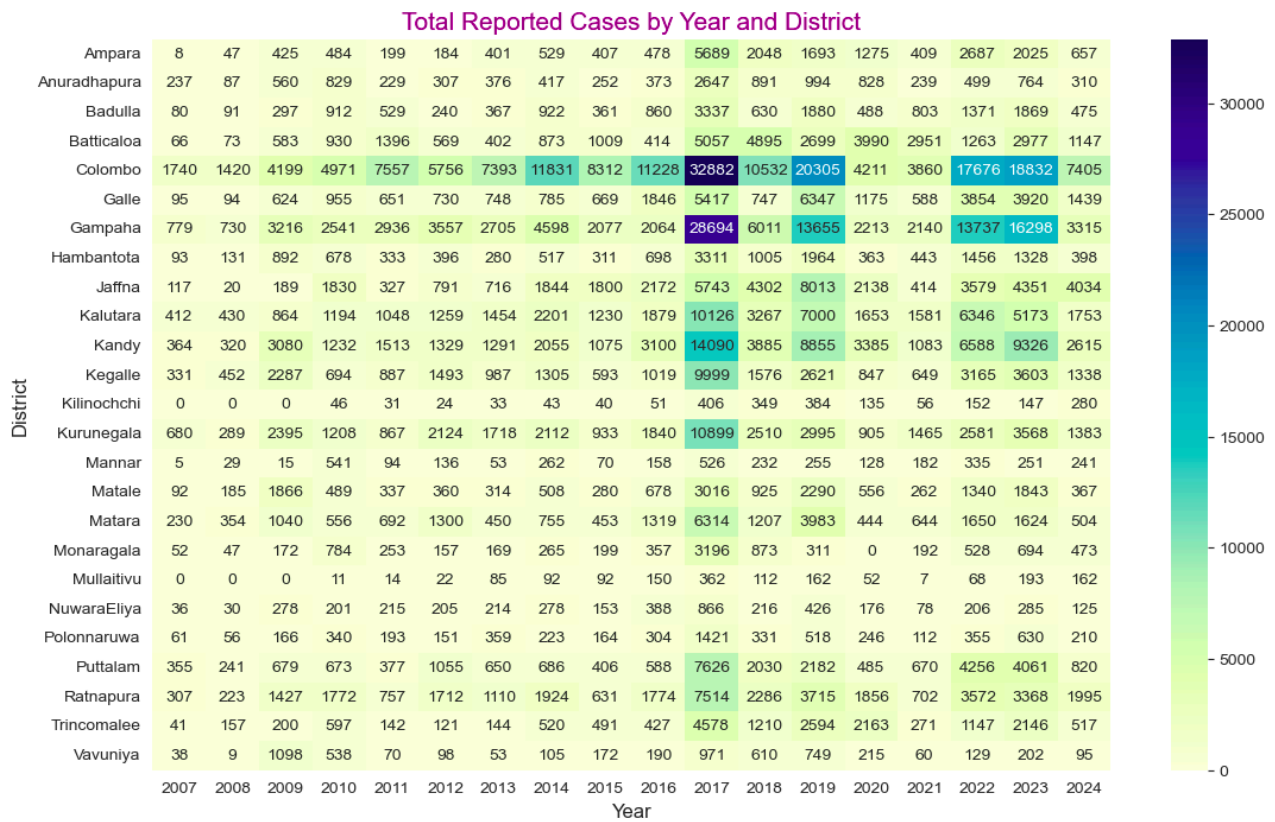
N.B

None of the climatic variables show a strong correlation with dengue case counts. The relationships are generally weak, indicating that other factors might play a more significant role in influencing dengue incidence. It is important to recognize that correlation does not imply causation. Other variables such as human behavior, mosquito control measures, and environmental conditions (e.g., stagnant water bodies from heavy rainfall) could be influencing the results.

Correlation Between Districts and Monthly Dengue Cases



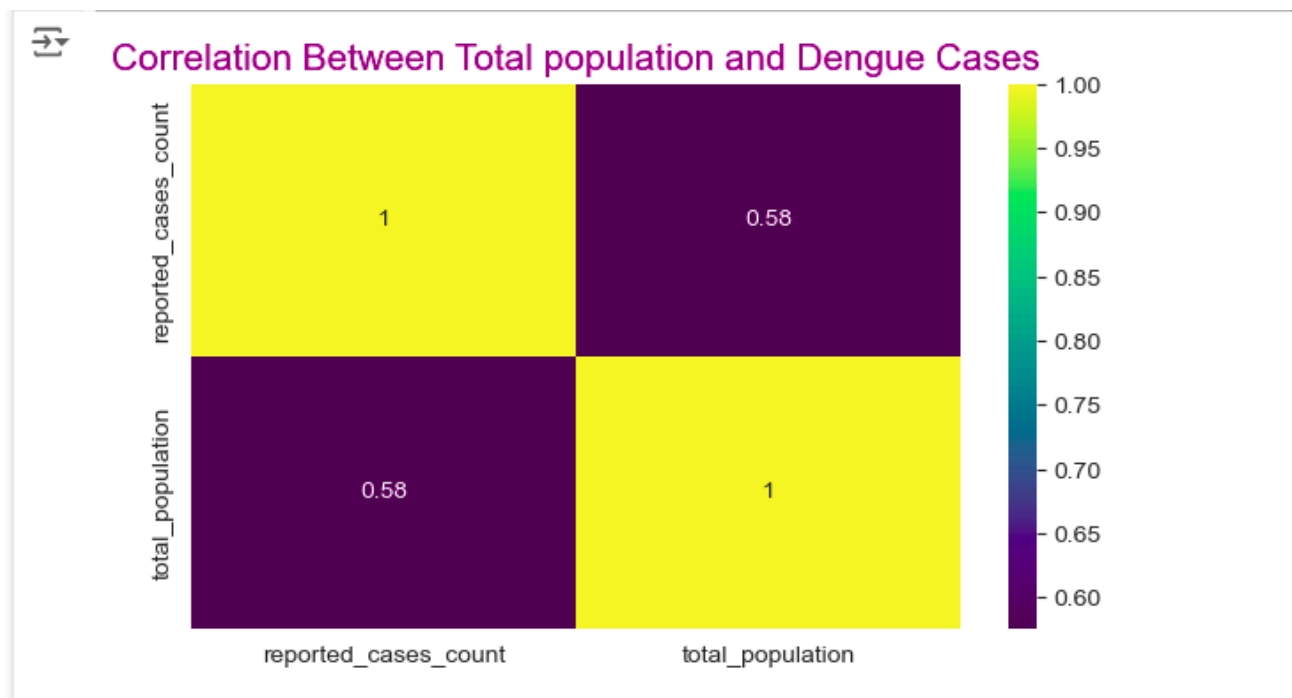
Colombo, Gampaha, and Kandy exhibit the highest number of reported dengue cases on a monthly basis. This strong correlation suggests that these districts experience a higher frequency of outbreaks compared to others. (Go to section 3.1, Outbreaks per districts)



Correlation Between Districts and Yearly Dengue Cases

The years **2017** and **2019** were particularly challenging for all districts in Sri Lanka, with 2017 being especially severe. During these years, nearly all districts experienced a significant spike in reported dengue cases. This aligns with section 5, High-level Annual Dengue Case Trends, which highlights 2017 as a peak year for dengue outbreaks across the country.

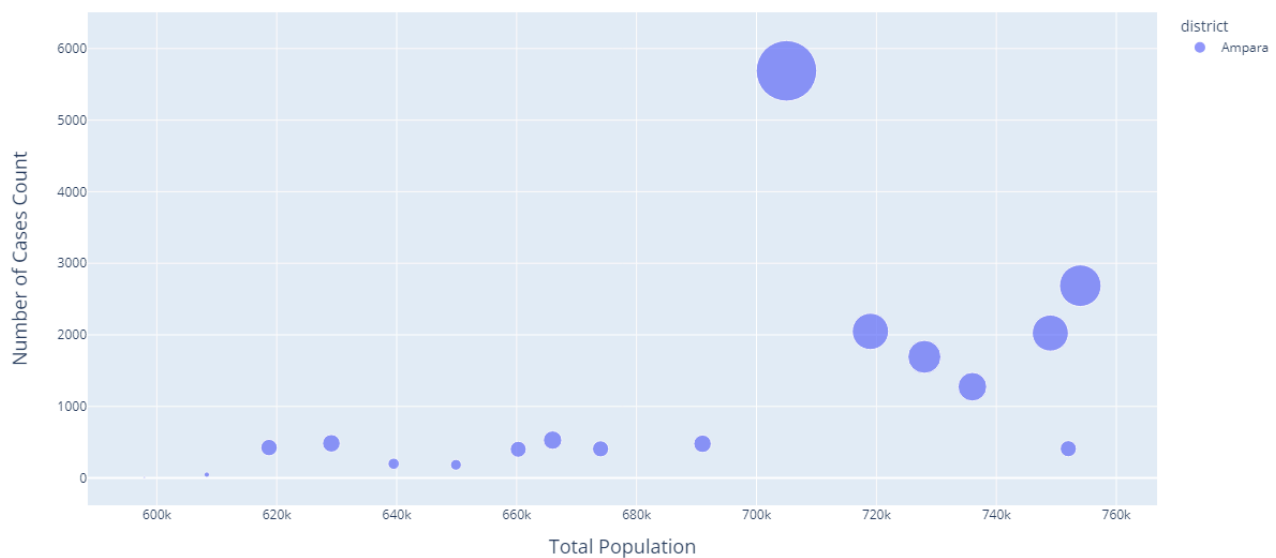
Correlation between Population and Reported Dengue Cases



A correlation coefficient of **0.58** indicates a moderately positive relationship between the total population and the number of reported dengue cases. This suggests that as the population increases, the number of dengue cases tends to rise as well. While the relationship is not perfectly linear, the correlation is significant enough to indicate a noticeable trend. This insight highlights the potential impact of population density on dengue transmission and the importance of monitoring densely populated areas for dengue outbreaks.

Correlation between Population and Reported Dengue Cases in Ampara District

Correlation between Population and Reported dengue cases in Ampara District

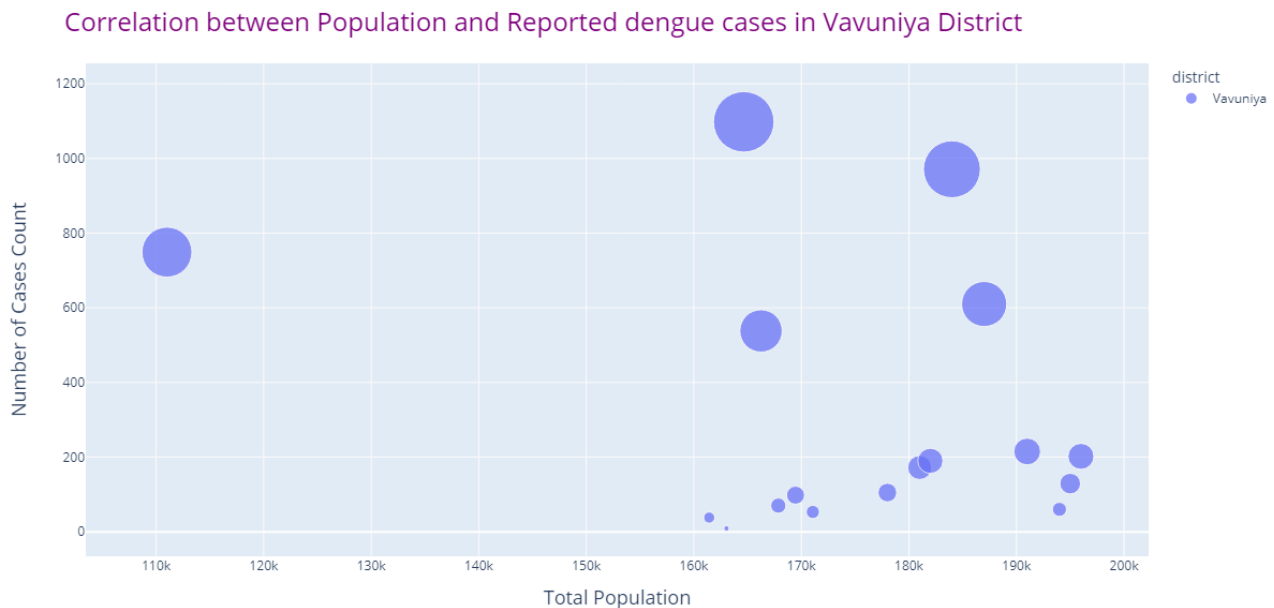


Correlation Analysis in Ampara District

In **Ampara**, there is a **moderate positive correlation** between the year and the number of reported dengue cases (**0.52**), and a **strong positive correlation** between the year and the total population (**0.99**). Additionally, a **moderate positive correlation** (**0.53**) is observed between the number of reported cases and the total population.

This analysis suggests that as the population in Ampara increases, there is a corresponding rise in the number of reported dengue cases. While the positive correlation between population growth and reported cases is moderate, it indicates a noticeable trend that points to the potential impact of population density on dengue transmission in the district.

Correlation between Population and Reported dengue cases in Vavuniya District



Correlation Analysis in Vavuniya District

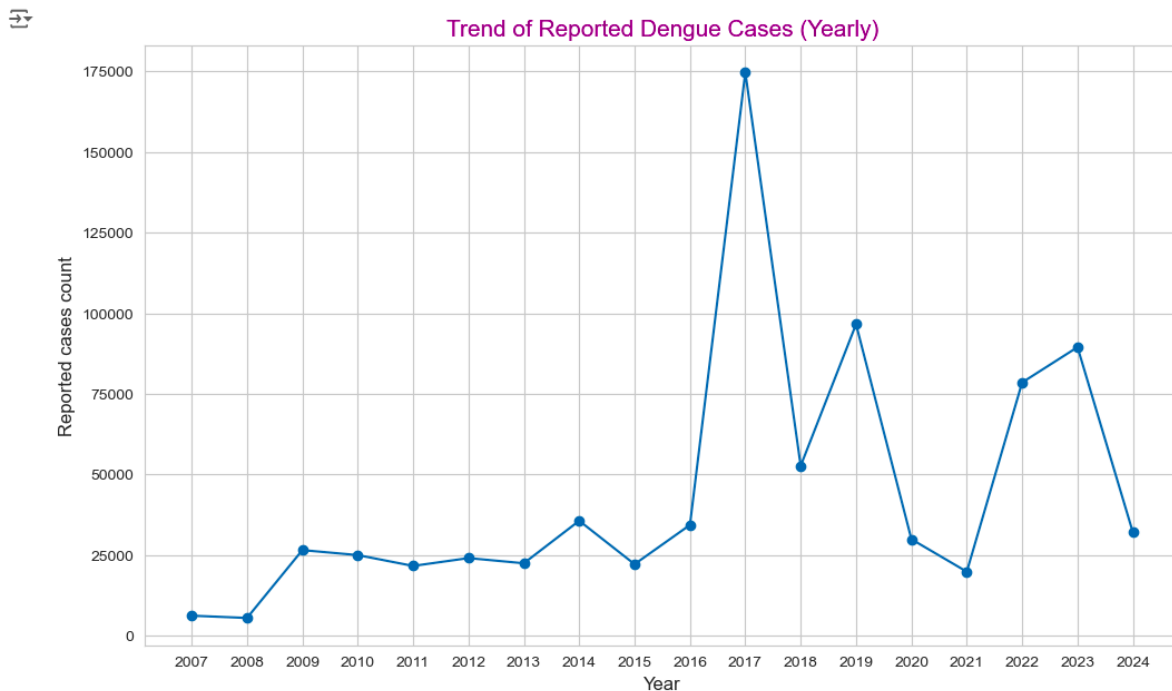
In **Vavuniya**, there is a **weak positive correlation** between the year and the number of reported dengue cases (**0.02**) and a **moderate positive correlation** between the year and the total population (**0.40**). However, a **weak negative correlation** (**-0.29**) is observed between the number of reported cases and the total population.

This suggests that as the population in Vavuniya increases, the number of reported dengue cases tends to decrease slightly. Although the relationship is weak, it indicates that population growth alone may not be driving dengue outbreaks in this district, and other factors may play a more significant role in case dynamics.

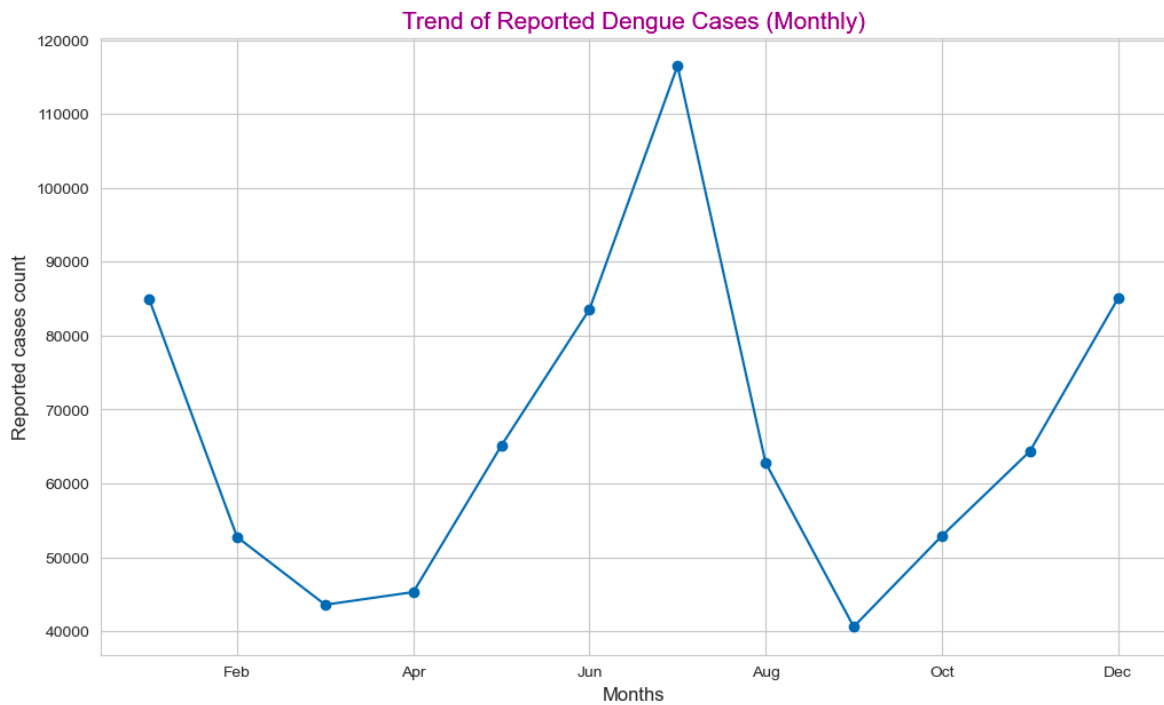
5. Visualization

- Graphs and Plots:
- Insights Derived from the Visuals:

1. High-level Annual Dengue Case Trends



The total number of reported cases reached its highest point in 2017 and its lowest point in 2008, as shown in the image above (Trend of Reported Dengue Cases (Yearly)). This variation highlights significant fluctuations in dengue activity over the years, as illustrated in the accompanying image.



Temporal Trends in Dengue Cases

The analysis of reported dengue cases over time reveals distinct seasonal patterns:

- **Increased Cases:** The number of reported cases rises from May to July and from October to December. This supports the hypothesis that dengue is endemic in Sri Lanka, with peak transmission typically occurring during the monsoon seasons.
- **Decreased Cases:** Conversely, the number of reported cases decreases from January to March and from July to September.

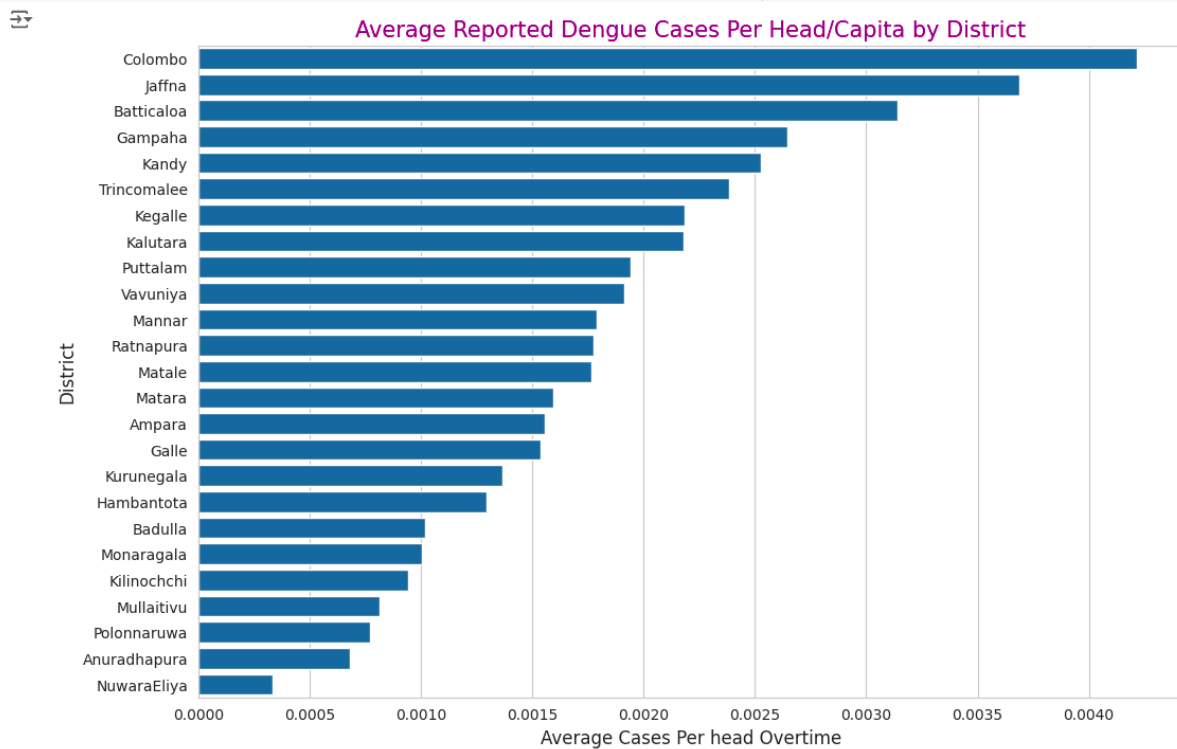
These patterns align with the project's assertion that seasonal monsoon periods are associated with higher dengue transmission, while other months experience lower case counts.

2. District-level Annual Dengue Case Trends

Total Number of Reported cases per district monthly and yearly trend

month	1	2	3	4	5	6	7	8	9	10	11	12
district												
Ampara	3171	1750	1319	1319	1952	2424	1264	677	582	905	1409	2873
Anuradhapura	1119	1009	738	638	842	1030	1527	610	432	526	717	1651
Badulla	1561	667	522	596	679	1138	2542	1308	947	1417	1911	2224
Batticaloa	4675	4000	3341	3637	3335	1947	1466	714	544	1182	2200	4253
Colombo	20120	12073	9142	9731	15069	21622	27446	13537	8450	11105	13316	18499
Galle	3080	1792	1518	1573	2711	3353	4571	2984	1789	2053	2489	2771
Gampaha	9649	5859	5637	6346	9849	12415	20167	9847	6184	7812	8438	9063
Hambantota	1253	1064	963	1134	1296	1594	2127	1219	875	930	1022	1120
Jaffna	7952	4154	2531	1929	1809	1816	1853	1322	1357	2936	5129	9592
Kalutara	4498	3044	2935	3084	4440	5461	7778	4116	2880	3163	3377	4094
Kandy	4620	2537	1925	2529	4765	7503	11312	6776	4598	6018	6402	6201
Kegalle	2332	1549	1537	1962	3674	4467	5840	3770	2120	2098	2073	2424
Kilinochchi	373	191	132	138	116	134	222	133	117	92	187	342
Kurunegala	4287	2368	2078	1891	3764	4535	6288	3474	2035	2721	3321	3710
Mannar	668	267	144	200	190	168	281	288	113	207	311	676
Matale	1263	842	677	652	914	1683	2372	1388	759	1413	1836	1909
Matara	2268	1467	1041	1167	1614	2142	4513	2560	1710	1828	1553	1656
Monaragala	773	560	461	522	626	924	1380	842	483	787	666	698
Mullaitivu	287	127	89	90	123	126	161	100	52	74	148	207
NuwaraEliya	410	268	221	206	274	457	815	428	235	296	334	432
Polonnaruwa	581	428	528	446	564	525	821	348	301	343	417	538
Puttalam	3854	1949	1229	1047	1669	1900	3025	1862	1294	2351	3364	4296
Ratnapura	2398	2019	1760	2032	3323	4895	7636	4004	2297	1972	2110	2199
Trincomalee	2990	2316	2761	2232	1325	999	760	304	225	360	966	2228
Vavuniya	820	441	356	194	203	296	347	184	229	303	656	1373

Effect of population on Reported Dengue cases per capita/head in different districts



Average Reported Dengue Cases Per Capita

The top 3 districts for Average Reported Dengue Cases Per Capita are Colombo, Jaffna, and Batticaloa.

6. Final Report and Presentation

- **Key Findings:**
- **Limitations of the Analysis:**
- **Recommendations and Next Steps:**

Prioritizing Interventions Based on Population and Case Burden

Since population shows the strongest **correlation** with the number of reported dengue cases over time (0.58), and the average reported cases per capita directly highlight districts with the highest case burden relative to their population, this provides a valuable basis for prioritizing interventions.

Using the data from the 'Average Reported Dengue Cases Per Capita by District Overtime' diagram, we can establish a recommendation order list to guide healthcare resource allocation. By focusing on districts with the highest case burden per capita, interventions can be more effectively targeted, ensuring that resources are deployed where the dengue impact is most significant.

Healthcare resources should be allocated to these districts based on this order, prioritizing areas with higher relative burdens, thus maximizing the efficiency of outbreak response and control efforts.